TABLE I
SUMMARY OF KEY NOTATIONS

Notation	Description
System Foundation	
λ, n, Δ	Security parameter, RLWE dimension, Scaling factor
$\mathbb{Z}_q, \mathcal{R}_q, \mathcal{R}_p$	Integer, ciphertext, and plaintext rings
φ, M	Ring isomorphism; Prime bounding modulus
χ_s, χ_e	Secret key and Gaussian error distributions
Client & Training Context	
$\mathcal{C}, N, K$	All client set, total count, selected count
$T, t, \mathcal{E}$	Total rounds, current round, local epochs
J, B, j	Total packs, PRPG seed, pack index
$R, sd_{i,j}$	Prefetch window; Pairwise PRG seed
$\mathcal{S}^{(t)}, \bar{\mathcal{S}}^{(t)}, \mathcal{P}^{(t^*)}$	Active, inactive, and prefetched client sets
S, s_i	Global decryption key; Local encryption key
Consensus & Model Updates	
$\mathcal{T}_i^{(t)}, \mathcal{K}^{(t)}$	Local Top-k indices; Consensus mask
$\mathbf{v}_i^{(t)}, v_{i,j}^{(t)}$	Vote bitmap; Binary voting element
$s_j^{(t)}, \theta, \tau_\theta$	Global score; Selection ratio; Truncation threshold
$\mathbf{W}^{(t)}, \mathbf{w}_i^{(t)}$	Global model; Local model
Gradients & Residuals	
$\mathbf{g}_i^{(t)}, \tilde{\mathbf{g}}_i^{(t)}, \mathbf{r}_i^{(t)}$	Raw gradient; Compensated gradient; Pack residual
$\mathrm{pt}_{i,j}^{(t)}, e_{i,j}^{(t)}$	Encoded gradient polynomial; Noise polynomial
γ_d, γ_s	Per-round and cross-round decay factors
Privacy & Cryptography	
$\mathbf{y}_{i,j}^{(t)}, \mathbf{z}_i^{(t)}, \hat{\mathbf{v}}_i^{(t)}$	Pairwise vector; Perturbation; Obfuscated vote
$\mathsf{ct}_{i,j}^{(t)}, \mathsf{ct}_{k,j}^{(t)}$	Local upload and compensatory ciphertexts
$C^{(t)}, C_j^{(t)}$	Global and pack-level aggregated ciphertexts; $C_j^{(t)}$ includes active gradients and KGC compensation
$\mathbf{\Delta}^{(t)}$	KGC compensatory perturbation
Prefetching & Aggregation	
$\mathcal{Q}$	Sliding queue of the latest R rounds' encrypted updates
$\mathcal{L}_{i,j}$	Client i's prefetch accumulator for pack j
$t_{\mathrm{last},i}$	Last training round of client i
$\mathbf{m}_{\mathrm{pre}}$	Prefetched plaintext from local accumulators
α, β	EMA decay and over-commitment factors
$\hat{d}^{(t)}, d^{(t)}$	EMA estimate and measured wall-clock round duration

Algorithm 1: ECPFL Main Procedure

Params.: $T, \mathcal{E}, \eta, R, k, M > N, \alpha, \beta$
Input: $\{\mathcal{D}_i\}_{i=1}^N$, $\mathbf{w}^{(0)}$
Output: Final global model

▷ Initialization
KGC generates private key shares $\{s_i\}$, global secret $S = \sum_i s_i$, PRPG seed B, and pairwise seeds $\{sd_{i,j}\}_{i<j}$
KGC sends (s_i, S), $\{sd_{i,j}\}_{j\neq i}$, and B to each client c_i
Server initializes EMA duration $\hat{d} \leftarrow d_0$
foreach $c_i \in \mathcal{C}$ **do**
  $\mathbf{w}_i \leftarrow \mathbf{w}^{(0)}$, $\mathbf{r}_i \leftarrow \mathbf{0}$, $t_{\mathrm{last},i} \leftarrow 0$, $\mathcal{L}_{i,j} \leftarrow \emptyset$ for all j
for $t = 1, 2, \ldots, T$ **do**
  ▷ Server-Side Execution
  Let target prefetch round $t^* = t + R$
  Select active and inactive sets $\mathcal{S}^{(t)}, \bar{\mathcal{S}}^{(t)}$
  $\hat{d} \leftarrow \alpha d^{(t-1)} + (1-\alpha)\hat{d}$ ▷ EMA round duration
  $\mathcal{P}^{(t^*)} \leftarrow$ sample $\lceil \beta K \rceil$ clients ▷ OC prefetch set
  KGC computes compensatory perturbation $\mathbf{\Delta}^{(t)}$ from $\bar{\mathcal{S}}^{(t)}$ and sends it to Server
  Receive obfuscated votes $\{\hat{\mathbf{v}}_i^{(t)}\}_{i\in\mathcal{S}^{(t)}}$ from active clients
  $\mathcal{K}^{(t)} \leftarrow$ ESP-CONSENSUS$(\{\hat{\mathbf{v}}_i^{(t)}\}_{i\in\mathcal{S}^{(t)}}, \mathbf{\Delta}^{(t)})$
  Broadcast consensus set $\mathcal{K}^{(t)}$ to active clients
  KGC generates compensatory ciphertexts $\{\mathsf{ct}_{k,j}^{(t)}\}$ from $(\bar{\mathcal{S}}^{(t)}, \mathcal{K}^{(t)})$ and sends them to Server
  Receive sparse ciphertexts $\{\mathsf{ct}_{i,j}^{(t)}\}$ from active clients
  $\{C_j^{(t)}\} \leftarrow$ ESP-AGG$(\mathcal{S}^{(t)}, \bar{\mathcal{S}}^{(t)}, \{\mathsf{ct}_{i,j}^{(t)}\}, \{\mathsf{ct}_{k,j}^{(t)}\}, \mathcal{K}^{(t)})$
  ECP-PREPAREPREFETCH$(\{C_j^{(t)}\}, \mathcal{K}^{(t)}, \mathcal{Q})$ ▷ Alg. 3
  ▷ Client-Side Execution
  for *each client* $c_i \in \mathcal{C}$ **do in parallel**
    if $c_i \in \mathcal{S}^{(t)}$ (Train) **then**
      if $t - t_{\mathrm{last},i} > 1$ **then**
        $\mathbf{w}_i \leftarrow$ ECP-APPLY$(\{\mathcal{L}_{i,j}\}, S, \mathbf{w}_i, B, \eta)$
      $\mathbf{w}_{i,\mathrm{start}} \leftarrow \mathbf{w}_i$
      $\mathbf{w}_i \leftarrow \mathsf{SGD}(\mathcal{D}_i, \mathbf{w}_i, \eta)$ for $\mathcal{E}$ epochs
      $\mathbf{g}_i^{(t)} \leftarrow \frac{1}{\eta}(\mathbf{w}_{i,\mathrm{start}} - \mathbf{w}_i)$
      $(\hat{\mathbf{v}}_i^{(t)}, \tilde{\mathbf{g}}_i^{(t)}) \leftarrow$ ESP-VOTE$(\mathbf{g}_i^{(t)}, \mathbf{r}_i, k, t)$
      Send vote $\hat{\mathbf{v}}_i^{(t)}$ to Server and wait for $\mathcal{K}^{(t)}$
      $a^{(t)} \leftarrow \mathsf{PRPG}(B, t)$; $gap_i \leftarrow t - t_{\mathrm{last},i}$
      $\{\mathsf{ct}_{i,j}^{(t)}\} \leftarrow$ ESP-ENC$(\tilde{\mathbf{g}}_i^{(t)}, \mathcal{K}^{(t)}, gap_i, s_i, a^{(t)})$
      Upload ciphertexts $\{\mathsf{ct}_{i,j}^{(t)}\}$ to Server; $t_{\mathrm{last},i} \leftarrow t$
    else if $c_i \in \mathcal{P}^{(t^*)}$ (Prefetch) **then**
      ECP-PREFETCH$(\mathcal{Q}, \{\mathcal{L}_{i,j}\}, t_{\mathrm{last},i})$ ▷ Alg. 3
  $d^{(t)} \leftarrow$ measured wall-clock duration of round t

I. KEY NOTATIONS

II. PERTURBATION-BASED CONSENSUS VOTING

A. Main and ESP Algorithms

III. EFFICIENT CIPHERTEXT PREFETCHING

As illustrated in Figure **??** and formalized in Algorithm 3, the ECP module synchronizes historically inactive clients before they rejoin training. The protocol operates in three phases.

a) Prepare Phase: At round t, the server presamples a prefetch client set $\mathcal{P}^{(t^*)}$ for a future target round $t^* = t + R$. Once ESP-AGG produces the pack-level aggregated ciphertexts $\{C_j^{(t)}\}_{j\in\mathcal{K}^{(t)}}$, the server pushes the tuple $(t, \mathcal{K}^{(t)}, \{C_j^{(t)}\})$ into a sliding queue $\mathcal{Q}$ and discards entries whose timestamps are older than $t - R + 1$. Thus $\mathcal{Q}$ always retains the encrypted updates from the latest R rounds. Each pack-level aggregate $C_j^{(\tau)}$ contains the active clients' encrypted gradients together with the KGC compensatory ciphertexts for the inactive clients; the compensation cancels the perturbation contributed by the inactive clients, so decrypting $C_j^{(\tau)}$ yields only the aggregated active gradient for pack j.

b) Prefetch Phase: During the interval from round t to round $t^* - 1$, each presampled client $i \in \mathcal{P}^{(t^*)}$ leverages its idle periods to incrementally download missed updates from $\mathcal{Q}$ into local accumulators $\{\mathcal{L}_{i,j}\}$. At any prefetch round t', the client scans $\mathcal{Q}$ and appends $(\tau, C_j^{(\tau)})$ to $\mathcal{L}_{i,j}$ for every uncollected round τ with $t_{\mathrm{last},i} < \tau \leq t'$ and every consensus pack $j \in \mathcal{K}^{(\tau)}$, skipping any round already present in $\mathcal{L}_{i,j}$. Any unfinished downloads are deferred to the next prefetch opportunity. Consequently, by round t^*, the client has accumulated all ciphertexts for rounds $t_{\mathrm{last},i}+1$ through t^*-1.

Algorithm 2: ESP Upstream Module

Params.: $J, k, M, \Delta, \gamma_d, \gamma_s$

Function ESP-VOTE($\mathbf{g}_i^{(t)}, \mathbf{r}_i, k, t$)
- $\tilde{\mathbf{g}}_i^{(t)} \leftarrow \mathbf{g}_i^{(t)} + \mathbf{r}_i$
- $\mathcal{T}_i \leftarrow \arg\max_{|\mathcal{T}|=k} \sum_{j\in\mathcal{T}} \|\tilde{\mathbf{g}}_{i,j}^{(t)}\|_2^2$
- $\mathbf{v}_i^{(t)} \in \{0,1\}^J$ with $v_{i,j}^{(t)} = 1$ if $j \in \mathcal{T}_i$
- $\mathbf{z}_i^{(t)} \leftarrow [\sum_{j>i} \mathsf{PRG}(sd_{i,j}, t) - \sum_{j<i} \mathsf{PRG}(sd_{j,i}, t)]_M$
- $\hat{\mathbf{v}}_i^{(t)} \leftarrow [\mathbf{v}_i^{(t)} + \mathbf{z}_i^{(t)}]_M$
- **return** $(\hat{\mathbf{v}}_i^{(t)}, \tilde{\mathbf{g}}_i^{(t)})$

Function ESP-CONSENSUS($\{\hat{\mathbf{v}}_i^{(t)}\}, \mathbf{\Delta}^{(t)}$)
- $\bar{\mathbf{v}}^{(t)} \leftarrow [\sum_{i\in\mathcal{S}^{(t)}} \hat{\mathbf{v}}_i^{(t)} + \mathbf{\Delta}^{(t)}]_M$
- $\mathcal{K}^{(t)} \leftarrow$ Top-k indices of $\bar{\mathbf{v}}^{(t)}$
- **return** $\mathcal{K}^{(t)}$

Function ESP-ENC($\tilde{\mathbf{g}}_i^{(t)}, \mathcal{K}^{(t)}, gap_i, s_i, a^{(t)}$)
- **foreach** $j \in \mathcal{K}^{(t)}$ **do**
  - $\mathrm{pt}_{i,j}^{(t)} \leftarrow \mathsf{CKKS.Encode}(\tilde{\mathbf{g}}_{i,j}^{(t)}, \Delta)$
  - $\mathsf{ct}_{i,j}^{(t)} \leftarrow [a^{(t)} s_i + \tilde{pe}_{i,j}^{(t)} + \mathrm{pt}_{i,j}^{(t)}]_q$
- $\mathbf{r}_i \leftarrow \gamma_d \gamma_s^{gap_i - 1} [\tilde{\mathbf{g}}_i^{(t)} \odot (\mathbf{1} - \mathcal{K}^{(t)})]$
- **return** $\{\mathsf{ct}_{i,j}^{(t)}\}_{j\in\mathcal{K}^{(t)}}$

Function ESP-AGG($\mathcal{S}^{(t)}, \bar{\mathcal{S}}^{(t)}, \{\mathsf{ct}_{i,j}^{(t)}\}, \{\mathsf{ct}_{k,j}^{(t)}\}, \mathcal{K}^{(t)}$)
- **foreach** $j \in \mathcal{K}^{(t)}$ **do**
  - $C_j^{(t)} \leftarrow \sum_{i\in\mathcal{S}^{(t)}} \mathsf{ct}_{i,j}^{(t)} + \sum_{k\in\bar{\mathcal{S}}^{(t)}} \mathsf{ct}_{k,j}^{(t)}$
- **return** $\{C_j^{(t)}\}_{j\in\mathcal{K}^{(t)}}$

c) Train Phase: When the client rejoins training at round t^* with $t^* - t_{\mathrm{last},i} > 1$, it first invokes ECP-APPLY before starting local training. For each pack j, it homomorphically aggregates the cached ciphertexts in $\mathcal{L}_{i,j}$ into a single composite and decrypts it with the global secret S. The recovered plaintext residuals are summed into $\mathbf{m}_{\mathrm{pre}}$, and the local model is updated as $\mathbf{w}_i \leftarrow \mathbf{w}_i - \eta\mathbf{m}_{\mathrm{pre}}$. Because ciphertext aggregation precedes decryption, the decryption cost is amortized over the prefetch horizon, and the incremental download traffic is spread across the idle interval rather than incurred all at once when the client rejoins.

To robustly manage system heterogeneity and client dropouts, ECPFL integrates the Over-Commitment (OC) and Exponential Moving Average (EMA) strategies established in FedFetch [**?**]. The server maintains an EMA round-duration estimate $\hat{d}^{(t)} = \alpha d^{(t-1)} + (1-\alpha)\hat{d}^{(t-1)}$ with decay factor α, and over-commits the prefetch set by sampling $|\mathcal{P}^{(t^*)}| = \lceil \beta K \rceil$ clients via the OC factor β. Guided by these schedules, bandwidth-limited clients initiate prefetching earlier to reduce severe communication latency, while faster clients remain idle longer to conserve local resources. Since $\mathcal{Q}$ retains only the latest R rounds, the downstream bandwidth for each prefetch client remains bounded by the ciphertext volume of R rounds regardless of the prefetch horizon.

Algorithm 3: ECP Download Module

Params.: Prefetch horizon R

Function ECP-PREPAREPREFETCH($C^{(t)}, \mathcal{K}^{(t)}, \mathcal{Q}$)
- Push $(t, \mathcal{K}^{(t)}, C^{(t)})$ into tail of $\mathcal{Q}$
- ▷ Retain only the latest R rounds
- **while** *oldest timestamp* $\tau_{\min} \in \mathcal{Q}$ *satisfies* $\tau_{\min} < t - R + 1$ **do**
  - discard oldest entry from $\mathcal{Q}$

Function ECP-PREFETCH($\mathcal{Q}, \{\mathcal{L}_{i,j}\}, t_{\mathrm{last},i}$)
- **foreach** $(\tau, \mathcal{K}^{(\tau)}, \{C_j^{(\tau)}\}) \in \mathcal{Q}$ *in increasing* τ **do**
  - **if** $\tau \le t_{\mathrm{last},i}$ **then** continue
  - **foreach** $j \in \mathcal{K}^{(\tau)}$ **do**
    - **if** $\tau \notin \mathcal{L}_{i,j}$ **then** append $(\tau, C_j^{(\tau)})$ to $\mathcal{L}_{i,j}$

Function ECP-APPLY($\{\mathcal{L}_{i,j}\}, S, \mathbf{w}_i, B, \eta$)
- $\mathbf{m}_{\mathrm{pre}} \leftarrow \mathbf{0}$
- **foreach** $j : \mathcal{L}_{i,j} \neq \emptyset$ **do**
  - $T_j \leftarrow \{\tau : (\tau, C_j^{(\tau)}) \in \mathcal{L}_{i,j}\}$
  - $a_j \leftarrow \sum_{\tau\in T_j} \mathsf{PRPG}(B, \tau)$
  - $C_j \leftarrow \sum_{(\tau, C_j^{(\tau)})\in\mathcal{L}_{i,j}} C_j^{(\tau)}$
  - $\mathbf{m}_{\mathrm{pre}} \leftarrow \mathbf{m}_{\mathrm{pre}} + \mathsf{HE.Dec}(a_j, S, C_j)$
- Clear $\{\mathcal{L}_{i,j}\}$
- **return** $\mathbf{w}_i - \eta\mathbf{m}_{\mathrm{pre}}$